Dimensionless Numbers

MATLAB Toolbox

Agustín Benítez Hernández*

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*agustin.benitez@umcc.cu, University of Matanzas, Cuba

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1 Dimensionless numbers

1.1 Reynolds number

Reynolds number (Pavlov et al., 1976, eq. 1-27):

$$Re = \frac{u\,L\,\rho}{\mu} = \frac{u\,L}{\nu} \tag{1}$$

where ρ is the density of the fluid (kg/m^3); μ is the dynamic viscosity of the fluid (Pa s) or ν is the kinematic viscosity of the fluid (m^2/s); u is the speed between the fluid and the solid (m/s); and L is a characteristic linear dimension (m). The characteristic linear dimension (L) can be:

- The hydraulic diameter of pipes [the inside diameter D_{in} if the pipe is circular, or equivalent diameter $D_{eq} = \frac{4\,Flow\;area}{Pipe\;perimeter}$ for non-circular pipes, (Pavlov et al., 1976, eqs. 1-30 and 4-17)].
- The hydraulic diameter for annular ducts (such as the outer channel in a tube-in-tube heat exchanger) is: $D_{annulus} = D_{in,o} - D_{out,i}$; where $D_{in,o}$ is the inside diameter of the outer pipe and $D_{out,i}$ is the outside diameter of the inner pipe.
- The distance between the plates for a fluid moving between two plane parallel surfaces (where the width is much greater than the space between the plates).

- Diameter of spherical particles in a fluid or equivalent diameter of non-spherical particles: $D_{eq} = 1.241\,\sqrt[3]{V_p} = 1.241\,\sqrt[3]{\frac{m_p}{\rho_p}}$
- Diameter of agitators (turbine or propeller) in cylindrical vessels stirred by a central rotating paddle.

For mass flow in circular tubes, the Reynolds number reduces to (Bergman et al., 2011, eq 8.6):

$$Re = \frac{4\,\dot{m}}{\pi\,D\,\mu} \tag{2}$$

where $\dot{m}$ is mass flow rate (kg/s) and D is the inside diameter of the pipe (m).

1.1.1 Reynolds in *Dimensionless MATLAB toolbox*

Re = `reynolds(` L, u, ρ, μ `)`

Re = `reynolds(` L, u, ν `)`

Re = `reynolds( 'Pipe',` D, $\dot{m}$, μ `)`

1.2 Froude number

Froude number (Pavlov et al., 1976, eq. 1-31):

$$Fr = \frac{u^2}{D\,g} \tag{3}$$

where u is the fluid speed (m/s); and D is the hydraulic diameter of pipes (m) (for details on equivalent diameters see section 1.1 on the preceding page).

1.2.1 Froude in *Dimensionless MATLAB toolbox*

Fr = `froude(` u, D `)`

Fr = `froude(` u, D, g `)`

1.3 Euler number

Euler number (Pavlov et al., 1976, eq. 1-32):

$$Eu = \frac{\Delta p}{\rho\, u^2} \tag{4}$$

where ρ is the density of the fluid (kg/m^3); u is the fluid speed (m/s) and $\Delta p = (p_{in} - p_{out})$ is the fluid pressure drop (Pa) between input and output of pipe.

1.3.1 Euler in *Dimensionless MATLAB toolbox*

Eu = `euler(` ρ, u `,` Δp `)`

Eu = `euler(` ρ, u `,` p_{in} `,` p_{out} `)`

Eu = `euler(` ρ, u `,` p_{out} `,` p_{in} `)`

1.4 Archimedes number

Archimedes number (Pavlov et al., 1976, eq. 3-3):

$$Ar = \frac{D^3\ (\rho_p - \rho)\,\rho\, g}{\mu^2} \tag{5}$$

where D is the diameter of spherical particles or equivalent diameter of non-spherical particles:

$$D = 1.241\,\sqrt[3]{V_p} = 1.241\,\sqrt[3]{\frac{m_p}{\rho_p}}$$

ρ is the density of the fluid (kg/m^3); μ is the dynamic viscosity of the fluid (Pa s); ρ_p is the density of the particle (kg/m^3); V_p and m_p are the volume (m^3) and mass (kg) of particle; and g is the gravitational acceleration ($g \approx 9.81$ m/s^2).

1.4.1 Archimedes in *Dimensionless MATLAB toolbox*

Ar = `archimedes(` D, ρ_p, ρ, μ `)`

Ar = `archimedes(` D, ρ_p, ρ, μ, g `)`

1.5 Lyashenko number

Lyashenko number (Pavlov et al., 1976, eq. 3-5):

$$Ly = \frac{u^3\,\rho^2}{\mu\,(\rho_p - \rho)\,g} \tag{6}$$

where u is the sedimentation velocity (m/s); ρ is the density of the fluid (kg/m^3); μ is the dynamic viscosity of the fluid (Pa s); ρ_p is the density of the particle (kg/m^3); and g is the gravitational acceleration ($g \approx 9.81$ m/s^2).

1.5.1 Lyashenko in *Dimensionless MATLAB toolbox*

Ly = `lyashenko(` u, ρ_p, ρ, μ `)`

Ly = `lyashenko(` u, ρ_p, ρ, μ, g `)`

2 Pavlov correlations

The Pavlov (1976, fig. 3-1) correlations are used to estimate diameter of spherical particles and sedimentation velocity of spherical particles:

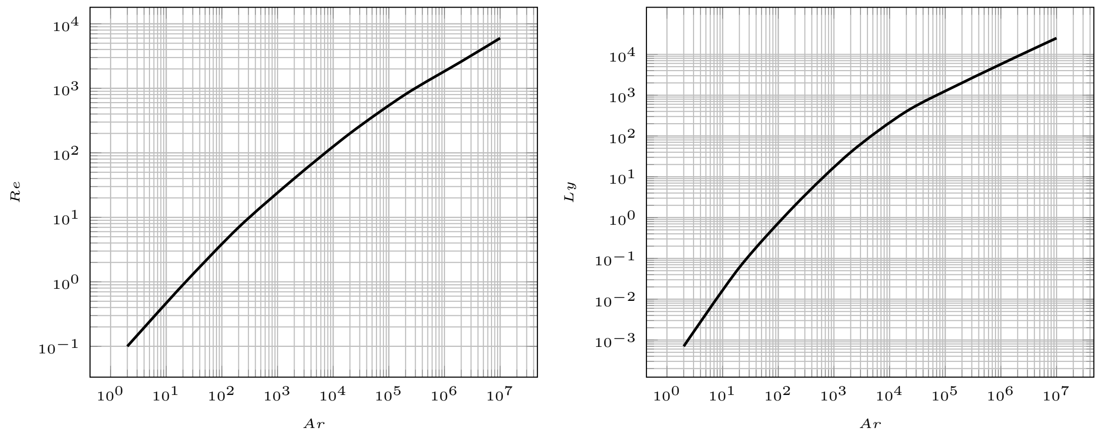

Figure 1: Pavlov correlations (Pavlov et al., 1976, fig. 3-1)

2.1 Pavlov correlations in *Dimensionless MATLAB toolbox*

Re = `pavlovcorr(` Ar `)` *%by default* $Re = f(Ar)$

Re = `pavlovcorr( 'Re_Ar',` Ar `)`

Re = `pavlovcorr( 'Re_Ly',` Ly `)`

Ly = `pavlovcorr( 'Ly_Re',` Re `)`

Ly = `pavlovcorr( 'Ly_Ar',` Ar `)`

Ar = `pavlovcorr( 'Ar_Re',` Re `)`

Ar = `pavlovcorr( 'Ar_Ly',` Ly `)`

2.2 Pavlov correlations examples

Example 1 (Pavlov et al., 1976, examp. 3-2)

Find the sedimentation velocity in water of a quartz spherical particle of 0.9 mm of diameter. The quartz density (ρ_p) is 2650 kg/m^3 and the water temperature, 20 °C. The water properties at 20°C are: density (ρ), 1000 kg/m^3 and dynamic viscosity (μ), 10^{-3} Pa s.

Solution: Calculate the Archimedes number, find the Reynolds number by Pavlov correlation and solve Reynolds equation to find the sedimentation velocity.

MATLAB solution:

```
function y = velocity( u )
    Dp = 0.9E-3; % m
    rhop = 2650; % kg/m³
    rho = 1000; % kg/m³
    mu = 1E-3; % kg/s·m
    if nargin == 0
        y = fzero( 'velocity', 0 );
    else
        Ar = archimedes( Dp, rhop, rho, mu );
        Re1 = pavlovcorr( Ar );
        Re2 = reynolds( Dp, u, rho, mu );
        y = Re1 - Re2;
    end

>> u = velocity

u = 0.1412 %m/s
```

Example 2 (Pavlov et al., 1976, examp. 3-3)

Calculate the size of the largest spherical chalk particles that are dragged by a rising water flow. The water velocity is 0.5 m/s. The chalk density (ρ_p) is 2710 kg/m^3 and the water temperature, 10 °C. The water properties at 10 °C are: density (ρ), 1000 kg/m^3 and dynamic viscosity (μ), 1.3 10^{-3} Pa s.

Solution: Calculate the Lyashenko number, find the Reynolds number by Pavlov correlation and solve Reynolds equation to find the diameter of particle.

MATLAB solution:

```
function y = diameter( Dp )
    u = 0.5; % m/s
    rhop = 2710; % kg/m³
    rho = 1000; % kg/m³
    mu = 1.3E-3; % kg/s·m
    if nargin == 0
        y = fzero( 'diameter', 0 );
    else
        Ly = lyashenko( u, rhop, rho, mu );
        Re1 = pavlovcorr( 'Re_Ly', Ly );
        Re2 = reynolds( Dp, u, rho, mu );
        y = Re1 - Re2;
    end

>> Dp = diameter

Dp = 0.005 %m
```

Example 3 (Pavlov et al., 1976, examp. 3-5-2)

Calculate the size of spherical shale particles that sediment at 0.1 m/s in water at 20 °C. The shale density (ρ_p) is 2200 kg/m^3. The water properties at 20 °C are: density (ρ), 1000 kg/m^3 and dynamic viscosity (μ), 10^{-3} Pa s.

Solution: Calculate the Lyashenko number, find the Archimedes number by Pavlov correlation and solve Archimedes equation to find the diameter of particle.

MATLAB solution:

```
function y = diameter( Dp )
    u = 0.1; % m/s
    rhop = 2200; % kg/m³
    rho = 1000; % kg/m³
    mu = 1E-3; % kg/s·m
    if nargin == 0
        y = fzero( 'diameter', 1 );
    else
        Ly = lyashenko( u, rhop, rho, mu );
        Ar1 = pavlovcorr( 'Ar_Ly', Ly );
        Ar2 = archimedes( Dp, rhop, rho, mu );
        y = Ar1 - Ar2;
    end

>> Dp = diameter

Dp = 7.5725E-4 %m
```

3 Convection correlations

3.1 Turbulent flow

In a fully developed turbulent flow, the main correlation between dimensionless numbers is:

$$Nu = f(Re, Pr)$$

where Nu is Nusselt number, Re is Reynolds number and Pr is Prandtl number. It can include other dimensionless groups as viscosity ratio or Darcy friction factor.

3.1.1 Circular tubes (pipes)

Colburn equation (Bergman et al., 2011, eq 8.59):

$$Nu = 0.023 Re^{\frac{4}{5}} Pr^{\frac{1}{3}} \tag{7}$$

Dittus-Boelter equation (Bergman et al., 2011, eq 8.60):

$$Nu = 0.023 Re^{\frac{4}{5}} Pr^{n} \tag{8}$$

where $n = 0.4$ for heating and $n = 0.3$ for cooling. Dittus-Boelter equation have been confirmed experimentally for the range $0.7 \leq Pr \leq 160$, $Re \gtrsim 10000$ and $\frac{L}{D} \geq 10$.

Sieder-Tate equation (Bergman et al., 2011, eq 8.61):

$$Nu = 0.027 Re^{\frac{4}{5}} Pr^{\frac{1}{3}} \left(\frac{\mu}{\mu_s}\right)^{0.14} \tag{9}$$

where μ is the viscosity at fluid temperature and μ_S the viscosity at surface temperature. These equation have been confirmed for $0.7 \leq Pr \leq 16700$, $Re \gtrsim 10000$ and $\frac{L}{D} \geq 10$.

Petukhov equation (Bergman et al., 2011, eq 8.62):

$$Nu = \frac{\frac{f}{8}\, Re\, Pr}{1.17 + 12.7 \left(\frac{f}{8}\right)^{\frac{1}{2}} \left(Pr^{\frac{2}{3}} - 1\right)} \tag{10}$$

where f is Darcy friction factor. The correlation is valid for $0.5 \leq Pr \leq 2000$ and $10^4 \leq Re \leq 5 \cdot 10^6$.

Gnielinski equation (Bergman et al., 2011, eq 8.63):

$$Nu = \frac{\frac{f}{8}\, (Re - 1000)\, Pr}{1 + 12.7 \left(\frac{f}{8}\right)^{\frac{1}{2}} \left(Pr^{\frac{2}{3}} - 1\right)} \tag{11}$$

These correlation is valid for $0.5 \leq Pr \leq 2000$ and $3000 \leq Re \leq 5 \cdot 10^6$.

3.1.2 Circular and non-circular pipes

Pavlov equation (Pavlov et al., 1976, eq 4-16):

$$Nu = 0.021\, \varepsilon\, Re^{0.8} Pr^{0.43} \left(\frac{Pr}{Pr_S}\right)^{0.25} \tag{12}$$

where Pr_S is the Prandtl number at surface temperature and ε is the correction coefficient:

Table 1: Correction coefficient ε

	Length/diameter ratio ($\frac{L}{D}$)				
Re	10	20	30	40	50 and more
$1 \cdot 10^4$	1.23	1.13	1.07	1.03	1
$2 \cdot 10^4$	1.18	1.10	1.05	1.02	1
$5 \cdot 10^4$	1.13	1.08	1.04	1.02	1
$1 \cdot 10^5$	1.10	1.06	1.03	1.02	1
$1 \cdot 10^6$	1.05	1.03	1.02	1.01	1

The diameter of pipe is the inside diameter (D_{in}) for circular pipes or equivalent diameter for non-circular pipes (see section 1.1 on page 2).

3.1.3 Convection correlations for turbulent flow in *Dimensionless MATLAB toolbox*

Nu = `nusseltcorr('Colburn',` Re, Pr `)`

Nu = `nusseltcorr('Dittus-Boelter heating',` Re, Pr `)`

Nu = `nusseltcorr('Dittus-Boelter cooling',` Re, Pr `)`

Nu = `nusseltcorr('Sieder-Tate',` Re, Pr, $\frac{\mu}{\mu_S}$ `)`

Nu = `nusseltcorr('Petukhov',` Re, Pr, f `)`

Nu = `nusseltcorr('Gnielinski',` Re, Pr, f `)`

Nu = `nusseltcorr('Pavlov',` Re, Pr, Pr_S, $\frac{L}{D}$ `)`

3.1.4 Turbulent flow convection example

Example 1 (Bergman et al., 2011, examp. 8.5)

Hot air flow with a mass rate ($\dot{m}$) of 0.05 kg/s through an uninsulated sheet metal duct of diameter (D) 0.15 m. The air properties are: Prandtl number (Pr), 0.70; thermal conductivity (k), 0.030 J/(s m °C) and dynamic viscosity (μ), $208 \cdot 10^{-7}$ Pa s. Find the convection heat transfer coefficient.

Solution: Calculate the Reynolds number, find the Nusselt number by Nusselt correlation using Dittus-Boelter equation for cooling and solve Nusselt equation to find the convection heat transfer coefficient.

MATLAB solution:

```
function y = coefficient( h )
    D = 0.15; %m
    m = 0.05; % kg/s
    mu = 208E-7; % Pa·m
    Pr = 0.70;
    k = 0.030; % W/m·°C
    if nargin == 0
        y = fzero( 'coefficient', 0 );
    else
        Re = reynolds( 'Pipe', D, m, mu );
        Nu1 = nusseltcorr( 'dittus-boelter cooling', Re, Pr );
        Nu2 = nusselt( h, k, D );
        y = Nu1 - Nu2;
    end

>> h = coefficient

h = 11.5896 % W/m²·°C
```

3.2 Free convection

In free convection the main dimensionless parameter is the Grashof number (Gr). It play the same role that Reynolds number in turbulent flow:

$$Nu = f(Gr, Pr)$$

Some correlations uses Rayleigh number (Ra), which is simply the product of Grashof and Prandtl numbers:

$$Nu = f(Ra, Pr)$$

3.2.1 Long horizontal cylinders (pipes)

Morgan equation (Bergman et al., 2011, eq 9.33):

$$Nu = C\, Ra^n \tag{13}$$

where:

Table 2: Morgan coefficients

Ra	C	n
10^{-10} to 10^{-2}	0.675	0.058
10^{-2} to 10^{2}	1.020	0.148
10^{2} to 10^{4}	0.850	0.188
10^{4} to 10^{7}	0.480	0.250
10^{7} to 10^{12}	0.125	0.333

Churchill-Chu equation (Bergman et al., 2011, eq 9.34):

$$Nu = \left\{ 0.60 + \frac{0.387 Ra^{\frac{1}{6}}}{\left[1 + \left(\frac{0.559}{Pr}\right)^{\frac{9}{16}}\right]^{\frac{8}{27}}} \right\}^2 \tag{14}$$

The Churchill-Chu correlation is valid for $Ra \lesssim 10^{12}$.

3.2.2 Vertical plates and vertical cylinders

Churchill-Chu equation for **vertical** objects (Bergman et al., 2011, eq 9.26):

$$Nu = \left\{ 0.825 + \frac{0.387 Ra^{\frac{1}{6}}}{\left[1 + \left(\frac{0.492}{Pr}\right)^{\frac{9}{16}}\right]^{\frac{8}{27}}} \right\}^2 \tag{15}$$

Churchill-Chu equation for **laminar** flow (Bergman et al., 2011, eq 9.27):

$$Nu = 0.68 + \frac{0.670 Ra^{\frac{1}{4}}}{\left[1 + \left(\frac{0.492}{Pr}\right)^{\frac{9}{16}}\right]^{\frac{4}{9}}} \tag{16}$$

These correlation is valid for $Ra \lesssim 10^{9}$.

Both correlations to vertical cylinders of height$L$, if the boundary layer thickness is much less than the cylinder diameter D. This condition is known to be satisfied when:

$$\frac{D}{L} \geq \frac{35}{Gr^{\frac{1}{4}}}$$

3.2.3 Horizontal and inclined plates

Equation for **upper surface of hot plate** or **lower surface of cold plate** (Bergman et al., 2011, eq 9.30 and 9.31):

$$Nu = \begin{cases} 0.54Ra^{\frac{1}{4}} & 10^4 \lesssim Ra \lesssim 10^7 \text{ and } Pr \geq 0.7 \\ 0.15Ra^{\frac{1}{3}} & 10^7 \lesssim Ra \lesssim 10^{11} \text{ and all } Pr \end{cases} \tag{17}$$

Equation for **lower surface of hot plate** or **upper surface of cold plate** (Bergman et al., 2011, eq 9.32):

$$Nu = 0.52Ra^{\frac{1}{5}} \tag{18}$$

valid for $10^4 \lesssim Ra \lesssim 10^9$ and $Pr \geq 0.7$.

Correlations 17 and 18 are recommended for inclined plates angles: $0 \leq \theta \leq 60^o$.

3.2.4 Convection correlations for free convection in *Dimensionless MATLAB toolbox*

```
Nu = nusseltcorr('Morgan', Ra )
Nu = nusseltcorr('Churchill-Chu', Ra, Pr )
Nu = nusseltcorr('Churchill-Chu vertical', Ra, Pr )
Nu = nusseltcorr('Churchill-Chu laminar', Ra, Pr )
Nu = nusseltcorr('upper hot', Ra )
Nu = nusseltcorr('upper cold', Ra )
Nu = nusseltcorr('lower hot', Ra )
Nu = nusseltcorr('lower cold', Ra )
```

3.2.5 Free convection example

Example 1 (Bergman et al., 2011, examp. 9.4)

A horizontal, high-pressure steam pipe of 0.1 m outside diameter (D) passes through a large room with air temperature (T_∞) 23 °C. The pipe has an outside surface temperature of (T_S) 165 °C.

Find the heat loss from pipe by free convection. The air property at $\bar{T}$ are: kinematic viscosity (ν), $22.8 \cdot 10^{-6}$ m^2/s; thermal diffusivity (α), $32.8 \cdot 10^{-6}$m^2/s; Prandtl number (Pr), 0.697; and thermal conductivity (k), 0.0313 J/(s m °C).

SOLUTION: Calculate the Rayleigh number; find the Nusselt number by Nusselt correlation using Churchill-Chu equation; solve Nusselt equation to find the convection heat transfer coefficient and calculate the heat loss per length unit:

$$q' = h\pi D \left(T_S - T_\infty\right)$$

MATLAB solution:

```
function y = qloss( h )
    Ts = 165; % °C
    Tinf = 23; % °C
    D = 0.1; % m
    nu = 22.8E-6; % m²/s
    alpha = 32.8E-6; % m²/s
    Pr = 0.697;
    k = 0.0313; % W/m·°C
    if nargin == 0
        h = fzero( 'qloss', 0 ); % W/m²·°C
        y = h * pi * D * (Ts - Tinf);
    else
        Ra = rayleigh( Ts, Tinf, D, nu, alpha );
        Nu1 = nusseltcorr( 'churchill-chu', Ra, Pr );
        Nu2 = nusselt( h, k, D );
        y = Nu1 - Nu2;
    end

>> q = qloss

q = 322.4703 % W/m
```

References

Bergman, T., Lavine, A., and Incropera, F. (2011). *Fundamentals of heat and mass transfer.* John Wiley & Sons, Inc., New York, 7th edition.

Pavlov, K. F., Romankov, P. G., and Noskov, A. A. (1976). *Примеры и задачи по курсу процессов* *и аппаратов химической технологии [Examples and problems to the course of unit operations of chemical engineering]. Химия* [Chemistry], Moscow.